BAYU FIRMANSYAH

Senior Full Stack Developer | Edutech Solution Architect

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PROFESSIONAL SUMMARY

Senior Full Stack Developer with deep expertise in engineering High-Scale Education Management Platforms. Currently driving core development for SiakadCloud and EdLink, serving 1,200+ universities in Indonesia. Proven track record in designing Complex Database Architectures for academic data, building Mobile-Ready RESTful APIs, and automating PDDIKTI Feeder reporting. Experienced in remote, project-based environments, offering a "Plug-and-Play" capability to build end-to-end academic systems from concept to deployment.

DOMAIN EXPERTISE (EDUTECH SPECIALIZATION)

- **Academic Core:** KRS (Study Plan) Logic, Grading Systems, GPA Calculation, Curriculum Management, Student Lifecycle (Admission to Graduation).
- **Compliance:** Automated Data Sync to PDDIKTI (Feeder Web Service).
- **Finance & Payment:** Tuitions management and Payment Gateway integration (Virtual Accounts).
- **LMS Features:** Digital Classroom, Assignment Submission, Quiz Logic, and Real-time Discussions.

TECHNICAL SKILLS

- **Backend & Architecture:** PHP (Laravel Expert), Node.js, Microservices Design, API-First Architecture (for Mobile/Web).
- **Frontend:** React.js, TypeScript, Next.js, Redux, Tailwind CSS (Pixel-Perfect Implementation).
- **Database Master:** PostgreSQL & MySQL (Complex Relational Schema Design for Academic Data, Query Optimization for High-Traffic/War-KRS).
- **DevOps & Remote Tools:** Docker, CI/CD Pipelines, Git, Jira, Slack (Async Communication).

PROFESSIONAL EXPERIENCE

PT Sentra Vidya Utama (SEVIMA) **Surabaya, Indonesia**
Senior Full Stack Developer (Product: SiakadCloud & EdLink) *Aug 2022 – Present*

- Leveraging 5 years of experience to build Indonesia’s leading Edutech SaaS.
- **Architected the "Core Academic Engine":** Designed and implemented the backend logic for SiakadCloud, handling complex academic scenarios (e.g., conflicting class schedules, credit limits based on GPA) for massive user bases.

- **Mobile-First API Development:** Built high-performance RESTful APIs specifically optimized for EdLink Mobile App (LMS), ensuring low-latency data delivery for students on Android/iOS devices.
- **High-Traffic Performance Tuning:** Solved the "KRS War" bottleneck (thousands of students accessing data simultaneously) by implementing Redis Caching and Database Indexing, improving response times by 40%.
- **PDDIKTI Compliance Module:** Developed a bridge module that automatically formats and pushes internal campus data to the National Higher Education Database (PDDIKTI), eliminating manual reporting errors.
- **Security Implementation:** Secured sensitive student data using JWT Authentication and Role-Based Access Control (RBAC) to ensure strict data privacy.

RevoU **Remote**

Vertical Mentor - Software Engineering *Aug 2023 – Jun 2024*

- Demonstrated strong Leadership & Communication by mentoring students to build full-stack applications from scratch, a skill critical for remote project coordination.
- Simplifying complex technical concepts for junior developers, ensuring smooth team collaboration.

ICUBE by SIRCLO **Yogyakarta, Indonesia**

Software Engineer *Mar 2022 – Jul 2022*

- **Complex Data Handling:** Managed real-time data synchronization for e-commerce, a skill directly transferable to managing real-time class enrollments and tuition payments.
- Developed high-fidelity Frontend using React.js, ensuring the platform is not only functional but visually professional.

EDUCATION

Bachelor of Science, Information Technology **2022 – Present**

Asia Cyber University

Diploma, Software Engineering **2019 – 2022**

SMK Negeri 4 Malang